## Instructions

- All questions carry equal weightage, except simulation questions, which have double the weightage.
- Copying will result in a loss of marks/penalty.
- All assumptions must be clearly stated along with appropriate figures (if any); otherwise, grading will not be done.
- Including the source code for simulation questions, along with the results, is mandatory. The source code must be properly commented, explaining each line.

# 1.Question

Two point charges of equal magnitude $q$ are positioned at $z = \pm d/2$. We aim to determine:

(a) The electric field everywhere on the $z$-axis.

(b) The electric field everywhere on the $x$-axis.

(c) The results of (a) and (b) if the charge at $z = -d/2$ is $-q$ instead of $+q$.

# Solution

### (a) Electric Field on the $z$-axis

Consider two point charges, both of magnitude $q$, located at $z = \pm d/2$. The electric field due to a point charge is given by:

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \tag{1}$$

For a point on the $z$-axis at $z$, the contributions from both charges are along the $z$-direction. The total field is:

$$E_z = \frac{1}{4\pi\epsilon_0} \left[ \frac{q}{(z - d/2)^2} - \frac{q}{(z + d/2)^2} \right]. \tag{2}$$

### (b) Electric Field on the $x$-axis

For a point on the $x$-axis at $(x, 0, 0)$, the distance to each charge is:

$$r_\pm = \sqrt{x^2 + (d/2)^2}. \tag{3}$$

The components of the electric field from each charge along the $z$-direction cancel, leaving only the $x$-component:

$$E_x = \frac{1}{4\pi\epsilon_0} \frac{2qx}{(x^2 + (d/2)^2)^{3/2}}. \tag{4}$$

### (c) Case When One Charge is $-q$

If the charge at $z = -d/2$ is $-q$, the fields add in the $z$-direction. The new field on the $z$-axis is:

$$E_z = \frac{1}{4\pi\epsilon_0}\left[\frac{q}{(z-d/2)^2} + \frac{q}{(z+d/2)^2}\right]. \tag{5}$$

For the $x$-axis, the $x$-components cancel, leaving only the $z$-component:

$$E_z = \frac{1}{4\pi\epsilon_0}\frac{2qd/2}{(x^2+(d/2)^2)^{3/2}}. \tag{6}$$

## 2.Question

A crude device for measuring charge consists of two small insulating spheres of radius $a$, one of which is fixed in position. The other is movable along the $x$-axis and is subject to a restraining force $kx$, where $k$ is a spring constant. The uncharged spheres are centered at $x = 0$ and $x = d$, the latter fixed. If the spheres are given equal and opposite charges of $Q/C$, obtain the expression by which $Q$ may be found as a function of $x$. Determine the maximum charge that can be measured in terms of $\epsilon_0$, $k$, and $d$, and state the separation of the spheres then. What happens if a larger charge is applied?

## Solution

A crude device for measuring charge consists of two small insulating spheres of radius $a$, one of which is fixed in position. The other is movable along the $x$ axis and is subject to a restraining force $kx$, where $k$ is a spring constant. The uncharged spheres are centered at $x = 0$ and $x = d$, the latter fixed. If the spheres are given equal and opposite charges of $Q$ coulombs:

(a) Obtain the expression by which $Q$ may be found as a function of $x$: The spheres will attract, and so the movable sphere at $x = 0$ will move toward the other until the spring and Coulomb forces balance. This will occur at location $x$ for the movable sphere. With equal and opposite forces, we have:

$$\frac{Q^2}{4\pi\epsilon_0(d-x)^2} = kx \tag{7}$$

from which:

$$Q = 2(d-x)\sqrt{\pi\epsilon_0 kx}. \tag{8}$$

(b) Determine the maximum charge that can be measured in terms of $\epsilon_0$, $k$, and $d$, and state the separation of the spheres then: With increasing charge, the spheres move toward each other until they just touch at $x_{\max} = d - 2a$. Using the part (a) result, we find the maximum measurable charge:

$$Q_{\max} = \frac{4a\sqrt{\pi\epsilon_0 k(d-2a)}}{.} \tag{9}$$

Presumably, some form of stop mechanism is placed at $x = x_{\max}$ to prevent the spheres from actually touching.

(c) What happens if a larger charge is applied? No further motion is possible, so nothing happens.

## 3. Question

A flux density field is given as

$$\mathbf{F}_1 = 5\mathbf{a}_z.$$

The task is to evaluate the outward flux of $\mathbf{F}_1$ through the hemispherical surface defined by $r = a$, $0 < \theta < \pi/2$, and $0 < \phi < 2\pi$. Next, consider what simple observation would have saved a lot of work in the previous part. Identifying symmetries or using alternative methods can simplify the calculations significantly. Now suppose the field is given by

$$\mathbf{F}_2 = 5z\mathbf{a}_z.$$

Using the appropriate surface integrals, determine the net outward flux of $\mathbf{F}_2$ through the closed surface consisting of the hemisphere from the first part and its circular base in the $xy$ plane. Finally, repeat the previous calculation by applying the divergence theorem and evaluating an appropriate volume integral. This approach should confirm the result obtained through direct surface integration.

## Solution

(a) A flux density field is given as $\mathbf{F}_1 = 5\mathbf{a}_z$. Evaluate the outward flux of $\mathbf{F}_1$ through the hemispherical surface, $r = a$, $0 < \theta < \pi/2$, $0 < \phi < 2\pi$.

The flux integral is

$$\Phi_1 = \int_{\text{hem.}} \mathbf{F}_1 \cdot d\mathbf{S} = \int_0^{2\pi} \int_0^{\pi/2} 5\mathbf{a}_z \cdot \mathbf{a}_r \underbrace{\cos\theta}_{\text{alignment}} a^2 \sin\theta \, d\theta \, d\phi$$

$$= -2\pi(5)a^2 \int_0^{\pi/2} \frac{\cos^2\theta}{2} d\theta = 5\pi a^2.$$

(b) What simple observation would have saved a lot of work in part (a)? The field is constant, and so the inward flux through the base of the hemisphere (of area $\pi a^2$) would be equal in magnitude to the outward flux through the upper surface. The flux through the base is a much easier calculation.

(c) Now suppose the field is given by $\mathbf{F}_2 = 5z\mathbf{a}_z$. Using the appropriate surface integrals, evaluate the net outward flux of $\mathbf{F}_2$ through the closed surface consisting of the hemisphere from part (a) and its circular base in the $xy$ plane.

Note that the integral over the base is zero since $\mathbf{F}_2 = 0$ there. The remaining flux integral is that over the hemisphere:

$$\Phi_2 = \int_0^{2\pi} \int_0^{\pi/2} 5z\mathbf{a}_z \cdot \mathbf{a}_r a^2 \sin\theta \, d\theta \, d\phi$$

$$= \int_0^{2\pi} \int_0^{\pi/2} 5(a\cos\theta)\cos\theta a^2 \sin\theta \, d\theta \, d\phi$$

$$= 10\pi a^3 \int_0^{\pi/2} \cos^2\theta \sin\theta \, d\theta$$

$$= -\frac{10}{3}\pi a^3 \cos^3\theta\Big|_0^{\pi/2} = \frac{10}{3}\pi a^3.$$

(d) Repeat part (c) by using the divergence theorem and an appropriate volume integral.

The divergence of $\mathbf{F}_2$ is just $dF_2/dz = 5$. We then integrate this over the hemisphere volume, which in this case involves just multiplying 5 by $(2/3)\pi a^3$, giving the same answer as in part (c).

## 4.Question

An infinitely long cylindrical dielectric of radius $b$ contains charge within its volume with a charge density given by

$$\rho_v = a\rho^2,$$

where $a$ is a constant. The goal is to determine the electric field strength $\mathbf{E}$ both inside and outside the cylinder.

## Solution

To solve this, we use Gauss's law in its differential form:

$$\nabla \cdot \mathbf{E} = \frac{\rho_v}{\epsilon_0}.$$

### Electric Field Inside the Cylinder ($\rho < b$)

Consider a cylindrical Gaussian surface of radius $\rho$ and length $L$. The total charge enclosed within this surface is obtained by integrating the charge density over the volume:

$$Q_{\text{enc}} = \int_0^{\rho} \int_0^{2\pi} \int_0^{L} (a\rho'^2)\rho' d\rho' d\phi dz.$$

Evaluating the integral:

$$Q_{\text{enc}} = \int_0^L dz \int_0^{2\pi} d\phi \int_0^{\rho} a\rho'^3 d\rho' = 2\pi a L \frac{\rho^4}{4} = \frac{2\pi a L \rho^4}{4}.$$

Using Gauss's law:

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0}.$$

Since the electric field is radially symmetric, we write:

$$E(\rho)(2\pi\rho L) = \frac{2\pi a L\rho^4}{4\epsilon_0}.$$

Solving for $E$:

$$E(\rho) = \frac{a\rho^3}{4\epsilon_0}, \quad \text{for } \rho < b.$$

### Electric Field Outside the Cylinder ($\rho > b$)

For $\rho > b$, the enclosed charge is the total charge within the entire cylinder, given by:

$$Q_{\text{total}} = \int_0^b \int_0^{2\pi} \int_0^L (a\rho'^2)\rho' d\rho' d\phi dz.$$

Evaluating the integral:

$$Q_{\text{total}} = 2\pi a L \frac{b^4}{4} = \frac{2\pi a L b^4}{4}.$$

Applying Gauss's law for $\rho > b$:

$$E(\rho)(2\pi\rho L) = \frac{2\pi a L b^4}{4\epsilon_0}.$$

Solving for $E$:

$$E(\rho) = \frac{ab^4}{4\epsilon_0\rho}, \quad \text{for } \rho > b.$$

## 5.Question

Given the vector field in cylindrical coordinates:

$$\vec{F} = \left[\frac{40}{s^2+1} + 3(\cos\phi + \sin\phi)\right]\hat{s} + 3(\cos\phi - \sin\phi)\hat{\phi} - 2\hat{z}$$

(a) Compute and plot the magnitude $|\vec{F}|$ as a function of $\phi$ for $s = 3$.
(b) Compute and plot the magnitude $|\vec{F}|$ as a function of $s$ for $\phi = 45°$.
(c) Calculate the divergence $\nabla \cdot \vec{F}$.
(d) Calculate the curl $\nabla \times \vec{F}$ and verify whether the field is conservative.

## Solution

### (a) $|\vec{F}|$ vs $\phi$ for $s = 3$

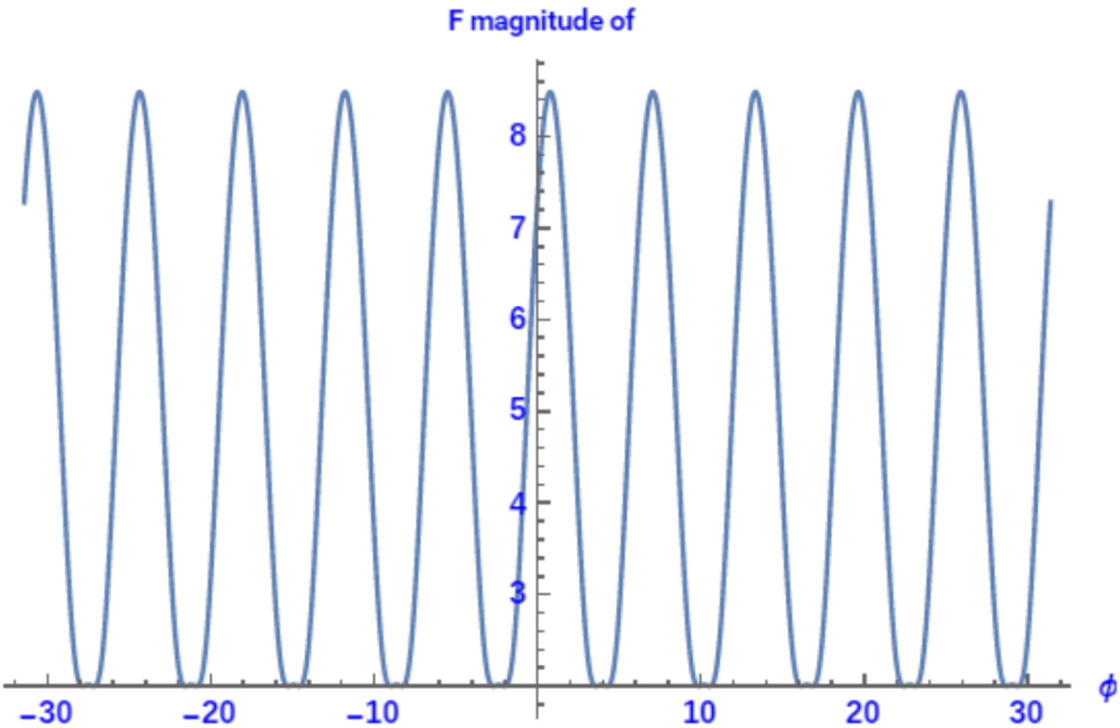
The norm of a vector in cylindrical coordinates is given by:

$$|\vec{F}| = \sqrt{(4 + 3(\cos\phi + \sin\phi))^2 + (3(\cos\phi - \sin\phi))^2 + (-2)^2} \tag{10}$$

Expanding,

$$|\vec{F}| = \sqrt{16 + 9(1 + 2\cos\phi\sin\phi) + 24(\cos\phi + \sin\phi) + 9(1 - 2\cos\phi\sin\phi) + 4} \tag{11}$$

$$|\vec{F}| = \sqrt{38 + 24(\cos\phi + \sin\phi)} \tag{12}$$




### (b) $|\vec{F}|$ vs $s$ for $\phi = 45°$

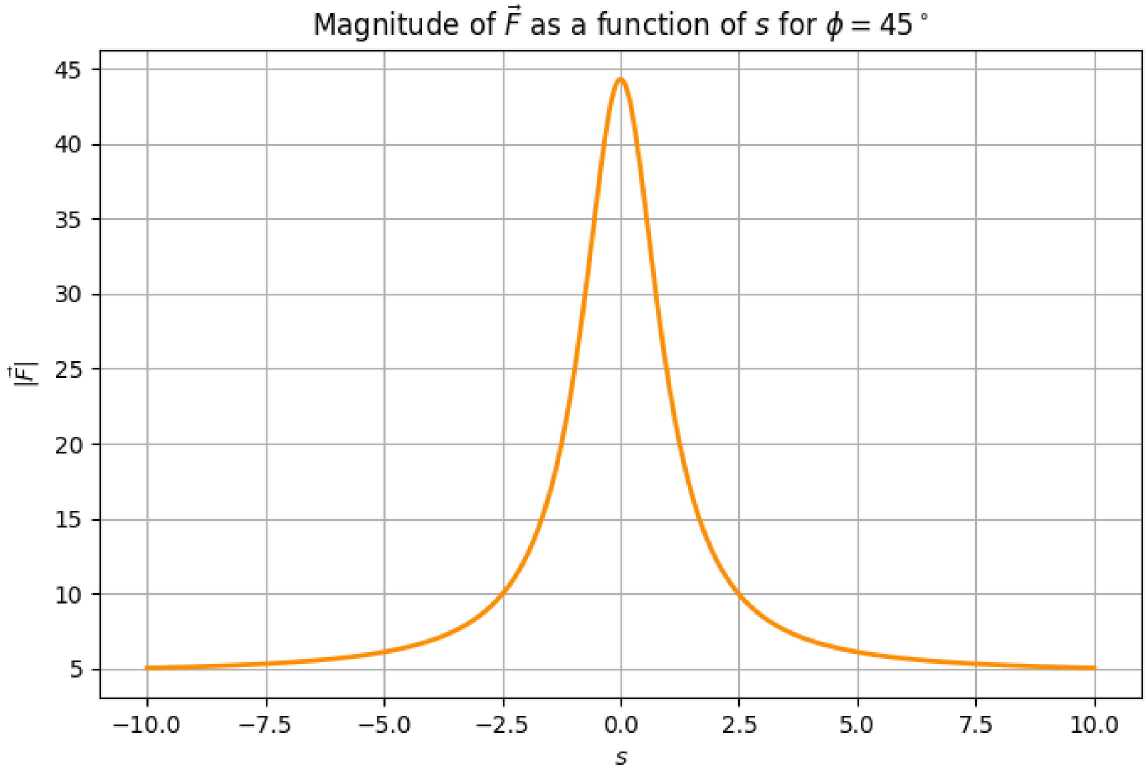
At $\phi = 45°$,

$$\vec{F} = \left[\frac{40}{s^2 + 1} + 3\sqrt{2}\right]\hat{s} - 2\hat{z} \tag{13}$$

The norm is given by:

$$|\vec{F}| = \sqrt{\left(\frac{40}{s^2+1} + 3\sqrt{2}\right)^2 + (-2)^2} \tag{14}$$

$$|\vec{F}| = \sqrt{\frac{1600}{(s^2+1)^2} + \frac{240\sqrt{2}}{s^2+1} + 2} \tag{15}$$




# Verification

## (a) Divergence of $\vec{F}$

The divergence in cylindrical coordinates is given by:

$$\nabla \cdot \vec{F} = \frac{1}{s}\frac{\partial}{\partial s}(sF_s) + \frac{1}{s}\frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z} \tag{16}$$

Substituting values,

$$\nabla \cdot \vec{F} = \frac{1}{s}\left(\frac{\partial}{\partial s}\left[s\left(\frac{40}{s^2+1} + 3(\cos\phi + \sin\phi)\right)\right]\right) + \frac{1}{s}\frac{\partial}{\partial \phi}\left[3(\cos\phi - \sin\phi)\right] + \frac{\partial}{\partial z}(-2) \tag{17}$$

$$\nabla \cdot \vec{F} = \frac{1}{s}\left(-\frac{40(s^2-1)}{(s^2+1)^2} + 3(\cos\phi + \sin\phi)\right) - \frac{3}{s}(\cos\phi + \sin\phi) \tag{18}$$

$$\nabla \cdot \vec{F} = -\frac{40(s^2-1)}{s(s^2+1)^2} \tag{19}$$

### (b) Curl of $\vec{F}$

The curl in cylindrical coordinates is given by:

$$\nabla \times \vec{F} = \frac{1}{s}\begin{vmatrix} \hat{s} & s\hat{\phi} & \hat{z} \\ \frac{\partial}{\partial s} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ F_s & sF_\phi & F_z \end{vmatrix} \tag{20}$$

Solving,

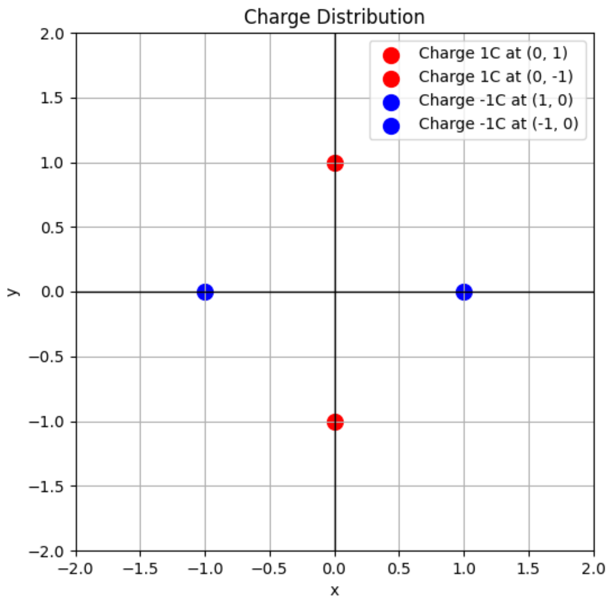
$$\nabla \times \vec{F} = \frac{1}{s}[(0)\hat{s} - (0)s\hat{\phi} + (3(\cos\phi - \sin\phi) - 3(\cos\phi - \sin\phi))\hat{z}] \tag{21}$$

$$\nabla \times \vec{F} = 0 \tag{22}$$

Thus, $\nabla \times \vec{F} = 0$, indicating that $\vec{F}$ is a conservative field.

## 6.Question

Refer to the charge distribution below;

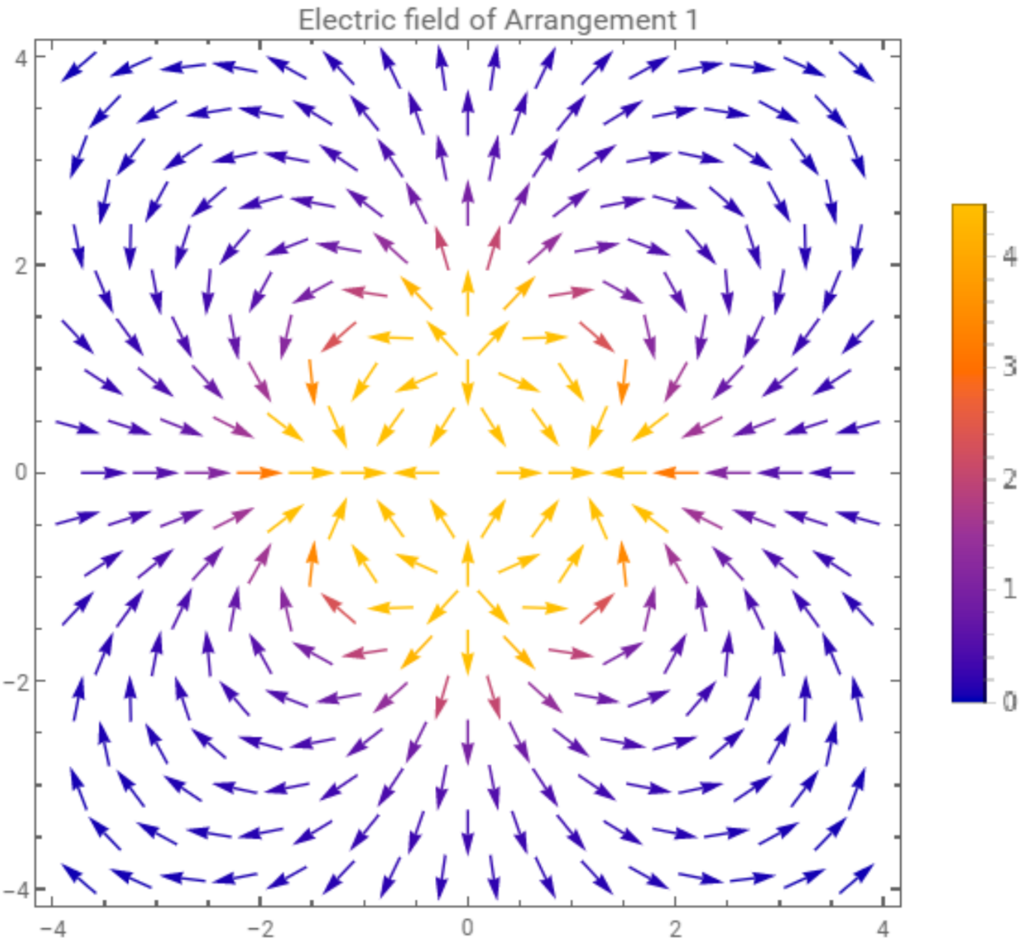



(a) Plot the Electric Field.
(b) Plot the potential with magnitude represented along the Z-axis.
(c) Compute the potential energy of the configuration.
(d) Use Gauss's Law to verify the divergence of the electric field.
(e) Verify whether the curl of the electric field is zero.

## Solution

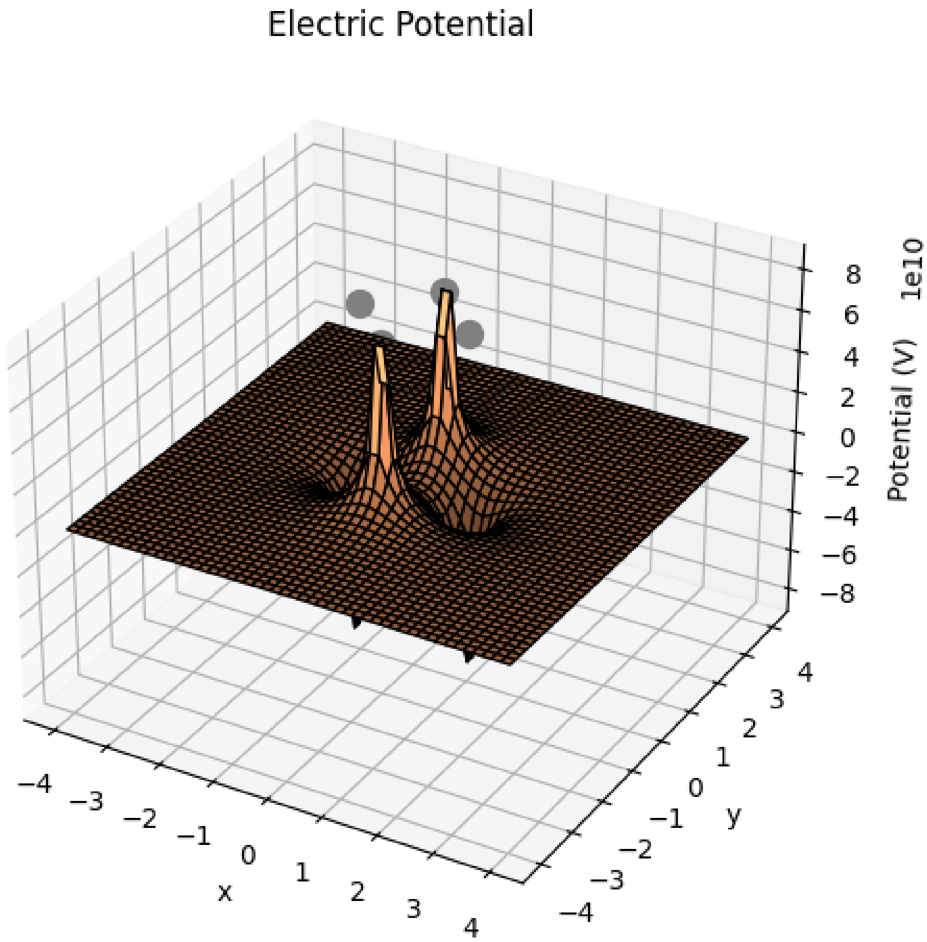
### (a) Electric Field Plot

The electric field has been plotted using Mathematica. The color plot represents the field intensity and the arrows indicate direction.

Out[11]=



### (b) Potential Plot

The potential function was plotted, with magnitude represented along the Z-axis. The contour lines indicate regions of equal potential.




### (c)Potential Energy

Potential energy of the configuration = Sum of potential energies between all possible combinations of charges

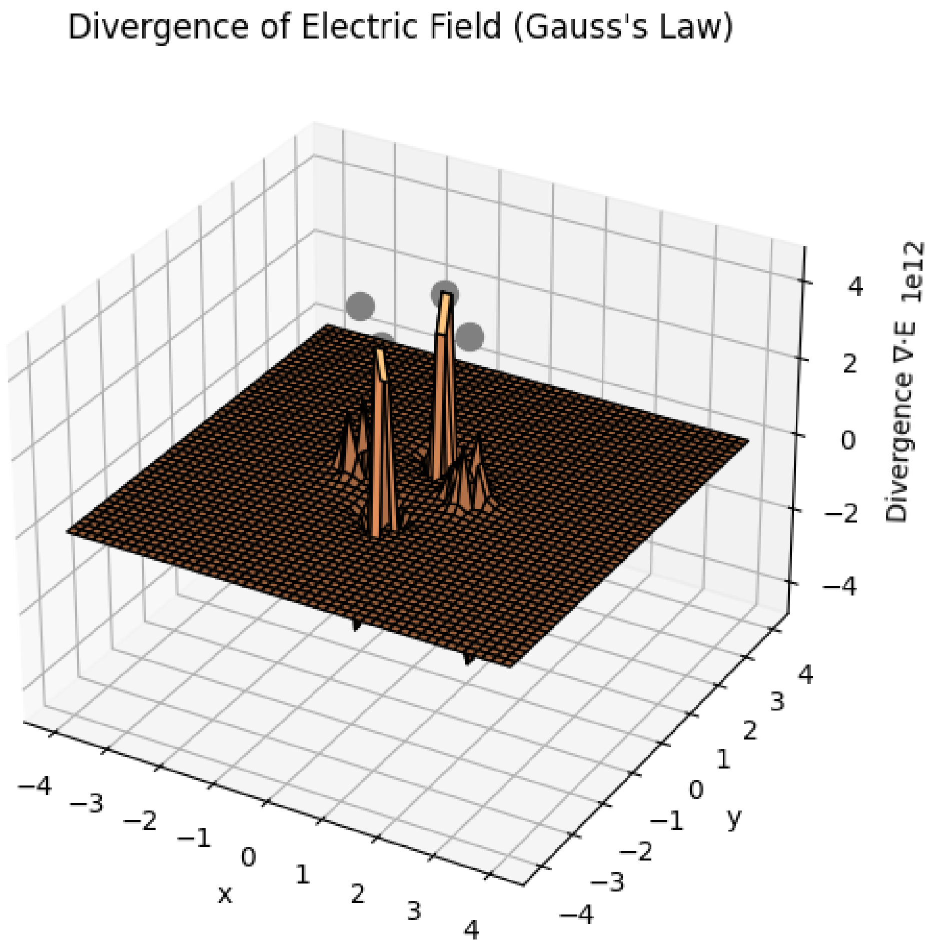
$$U = \frac{1}{4\pi\varepsilon_0}\left(4 \times \frac{(-1)(1)}{\sqrt{2}} + 2 \times \frac{1}{2}\right) = \left(-2\sqrt{2}+1\right)\frac{1}{4\pi\varepsilon_0}$$

### (d) Verification Using Gauss's Law

From Gauss's Law,

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \tag{23}$$

The computed divergence shows peaks at charge locations, confirming the presence of localized charge density.



### (e) Curl of the Electric Field

From Maxwell's equation,

$$\nabla \times \vec{E} = 0 \tag{24}$$

The computed curl is zero, confirming that the electric field is conservative.

## 7.Question

Given the spherically symmetric potential field in free space, $V(r) = V_0 e^{-r/a}$, determine the following: (a) Find the charge density $\rho_v$ at $r = a$. (b) Calculate the electric field $\mathbf{E}$ at $r = a$. (c) Compute the total charge.

## Solution

**Given the spherically-symmetric potential field in free space,**

$$V = V_0 e^{-r/a}$$

find:

### (a) Charge density $\rho_v$ at $r = a$

Using Poisson's equation,

$$\nabla^2 V = -\frac{\rho_v}{\epsilon_0}$$

which in this case becomes

$$-\frac{\rho_v}{\epsilon_0} = \frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dV}{dr}\right) = \frac{-V_0}{ar^2}\frac{d}{dr}\left(r^2 e^{-r/a}\right)$$
$$= \frac{-V_0}{ar}\left(2 - \frac{r}{a}\right) e^{-r/a}$$

From which,

$$\rho_v(r) = \frac{\epsilon_0 V_0}{ar}\left(2 - \frac{r}{a}\right) e^{-r/a}$$

At $r = a$:

$$\rho_v(a) = \frac{\epsilon_0 V_0}{a^2} e^{-1} \quad \mathrm{C/m^3}$$

### (b) Electric field at $r = a$

The electric field is found through the negative gradient:

$$\mathbf{E}(r) = -\nabla V = -\frac{dV}{dr}\mathbf{a_r} = \frac{V_0}{a}e^{-r/a}\mathbf{a_r}$$

At $r = a$:

$$\mathbf{E}(a) = \frac{V_0}{a}e^{-1}\mathbf{a_r} \quad \text{V/m}$$

### (c) Total charge

The easiest way is to first find the electric flux density, which from part (b) is:

$$\mathbf{D} = \epsilon_0\mathbf{E} = (\epsilon_0 V_0/a)e^{-r/a}\mathbf{a_r}$$

Then the net outward flux of $\mathbf{D}$ through a sphere of radius $r$ would be:

$$\Phi(r) = Q_{\text{encl}}(r) = \oint_S \mathbf{D} \cdot d\mathbf{S} = 4\pi r^2 D$$

$$= 4\pi\epsilon_0 V_0 r^2 e^{-r/a} \quad \text{C}$$

As $r \to \infty$, this result approaches zero, so the total charge is therefore:

$$Q_{\text{net}} = 0.$$

## 8.Question

A parallel-plate capacitor has plates located at $z = 0$ and $z = d$. The region between the plates is filled with a material that contains a uniform volume charge density $\rho_0$ C/m$^3$ and has permittivity $\epsilon$. Both plates are held at ground potential.

(a) Determine the potential field between the plates.

(b) Determine the electric field intensity $\mathbf{E}$ between the plates.

(c) Repeat parts (a) and (b) for the case where the plate at $z = d$ is raised to a potential $V_0$, with the plate at $z = 0$ grounded.

## Solution

A parallel-plate capacitor has plates located at $z = 0$ and $z = d$. The region between plates is filled with a material containing volume charge of uniform density $\rho_0$ C/m$^3$, and which has permittivity $\epsilon$. Both plates are held at ground potential.

## (a) Potential Field Between Plates

We solve Poisson's equation, under the assumption that $V$ varies only with $z$, also assume homogeneous, isotropic dielectric:

$$\nabla^2 V = \frac{d^2V}{dz^2} = -\frac{\rho_0}{\epsilon}$$

Solving,

$$V = \frac{-\rho_0 z^2}{2\epsilon} + C_1 z + C_2$$

At $z = 0$, $V = 0$, so $C_2 = 0$. Then, at $z = d$, $V = 0$ as well, so we find

$$C_1 = \frac{\rho_0 d}{2\epsilon}.$$

Thus, the potential field is:

$$V(z) = \frac{\rho_0 z}{2\epsilon}(d - z).$$

## (b) Electric Field Between Plates

The electric field intensity $\mathbf{E}$ is found using:

$$\mathbf{E} = -\nabla V = -\frac{dV}{dz}\mathbf{a_z}$$

$$= -\frac{d}{dz}\left[\frac{\rho_0 z}{2\epsilon}(d - z)\right] = \frac{\rho_0}{2\epsilon}(2z - d)\mathbf{a_z} \quad \text{V/m.}$$

## (c) Case with Plate at $z = d$ Raised to Potential $V_0$

If the plate at $z = d$ is raised to potential $V_0$ while the $z = 0$ plate remains grounded, we start with:

$$V(z) = \frac{-\rho_0 z^2}{2\epsilon} + C_1 z + C_2.$$

Since $V(0) = 0$, we again set $C_2 = 0$. At $z = d$,

$$V(z = d) = V_0 = \frac{-\rho_0 d^2}{2\epsilon} + C_1 d.$$

Solving for $C_1$,

$$C_1 = \frac{V_0}{d} + \frac{\rho_0 d}{2\epsilon}.$$

Thus, the new potential field is:

$$V(z) = \frac{V_0}{d}z + \frac{\rho_0 z}{2\epsilon}(d - z).$$

The electric field is:

$$\mathbf{E} = -\frac{dV}{dz}\mathbf{a_z} = -\frac{V_0}{d}\mathbf{a_z} + \frac{\rho_0}{2\epsilon}(2z - d)\mathbf{a_z} \quad \text{V/m}.$$